

Logic Gates

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George Boole has written a book "An investigation of the laws of thought" this book is based on logic and Probabilities.

Later on Claude E. Shannon at Massachusetts Institute of Technology wrote a thesis. Today Boolean Algebra is the backbone of Computer Circuit Analysis.

1 and 0
High and low
True and False
On and Off

Fundamental concept of boolean algebra

Only two mathematical operations

addition

Multiplication (Repeated addition)

OR

AND

Conjunction (Logical Addition)

Disjunction (Logical Multiplication)

The '+' symbol is read as X or Y

$X + Y$

$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 0 = 1$$

$$1 + 1 = 1$$

$X \cdot Y$

$$0 \cdot 0 = 0$$

$$1 \cdot 0 = 0$$

$$0 \cdot 1 = 0$$

$$1 \cdot 1 = 1$$

$a > b$ & $a > c$ Then a is greatest

Three basic laws of Boolean Algebra are same as normal algebra:

1. Commutative law $\rightarrow A + B = B + A, \quad A \cdot B = B \cdot A$

2. Associative law $\rightarrow (A + B) + C = A + (B + C), \quad A(B \cdot C) = (A \cdot B) \cdot C$

3. Distributive law $\rightarrow A(B + C) = AB + AC$

$$A + BC = (A + B)(A + C)$$

Logic Gates: A logic gate is defined as an electronic circuit with two or more input signals and it depicts the logic clearly.

Three types of Logic Gates:

1. Basic Gates

2. Universal Gates

3. Exclusive Gates

$\rightarrow$ OR gate
 $\rightarrow$ AND gate
 $\rightarrow$ NOT gate

$\rightarrow$ NOR gate
 $\rightarrow$ NAND gate

$\rightarrow$ XOR gate
 $\rightarrow$ XNOR gate

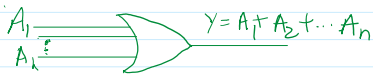
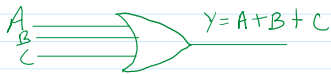
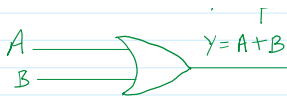
BASIC GATES:

(i) OR Gate:- An OR gate is a logic ckt. with 2 or more input signals and '1' output signal. The output signal will be high or '1', if any of the input signal is high or '1', or true or on (logic-1).

OR gate performs logical Addition. It is represented by '+' plus symbol.

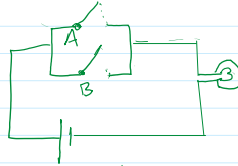


A	B	Y
0	0	0



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

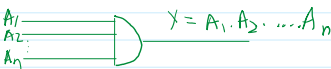
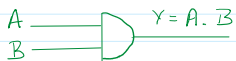
Truth table of OR gate.



Electrical representation of OR gate.

ii) AND Gate: An AND gate is a logic ckt. with 2 or more input signals and '1' output signal. The output signal will be high or '1', if all of the input signal is high or '1', or true or on (logic-1).

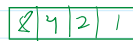
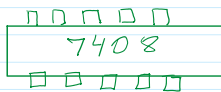
AND gate performs logical Multiplication. It is represented by 'dot' symbol.



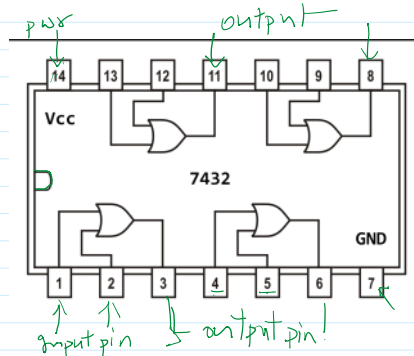
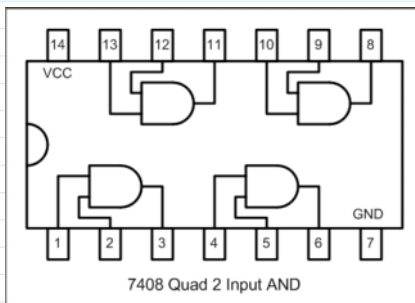
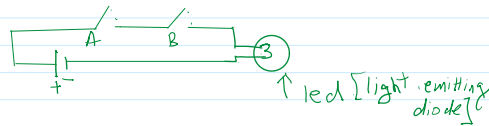
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Truth table of AND gate.

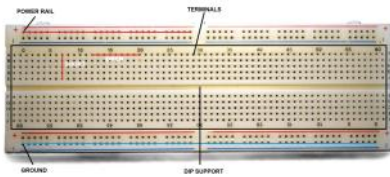
Integrated circuit (IC)



← quad input → Registers
→ capacitor
→ diodes.
← 16 different combination



Bread board



NOT Gate $A \rightarrow \neg A$

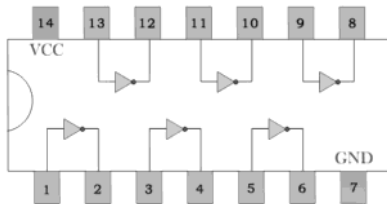
An inverter or NOT gate is the ckt. that takes one input signal and output signal is always opposite or complement of the input signal.

A	$\bar{A}$
0	1
1	0

Truth table for NOT gate.



Inverter ckt.
or
short ckt.



7404 Six Inverter
NOT gate IC.

BASIC DUALITY IN BOOLEAN ALGEBRA

We state the duality theorem in the following

1. changing each OR (+) sign to an AND (·) sign.
2. changing each AND (·) sign to an OR (+) sign
3. Complement each '0' to '1' or '1' to '0'

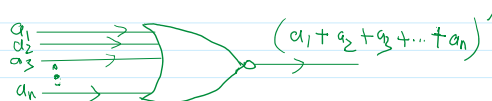
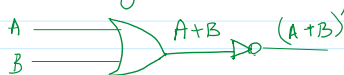
$$A + 0 = A \cdot 1 = A$$

dual relationship $\left\{ \begin{array}{l} A \cdot (B + C) = AB + AC \\ A + B \cdot C = (A + B)(A + C) \end{array} \right\}$ distributive law

dual of $\rightarrow (1 + A) = 0 \cdot A = 0$
 $(1 + \bar{A}) = 0 \cdot \bar{A} = 0$

Universal Gates (De-Morgan's law)

1. NOR: This gate takes two or more input and output is inverter of OR gate.

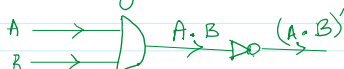


A	B	$(A+B)'$
0	0	1
0	1	0
1	0	0
1	1	0

Truth table.

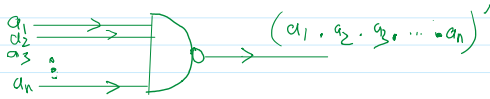
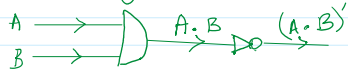
$$(A+B)' = \bar{A} \cdot \bar{B}$$

2. NAND: This gate takes two or more input and output is inverter of AND gate.



A	B	$(A \cdot B)'$
0	0	1

2. **NAND** : This gate takes two or more input and output is inverse of AND gate.



A	B	(A.B)'
0	0	1
0	1	1
1	0	1
1	1	0

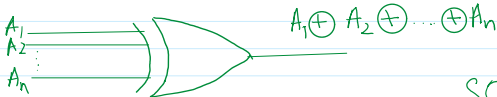
Truth table

$$(A.B)' = \bar{A} + \bar{B}$$

EXCLUSIVE GATES

1. **XOR Gate** : It takes 2 or more input signals and output one signal. The output signal is high or 1 or true, iff we had odd number of high or 1 or true input signals, otherwise output is low or 0.

It is represented by '+' plus in a circle symbol.

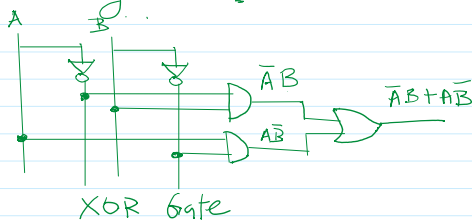


A	B	A ⊕ B
0	0	0
0	1	1
1	0	1
1	1	0

$A+B \leftarrow$
 $\checkmark \bar{A}B$
 $\checkmark A\bar{B}$
 $\bar{A}+\bar{B} \leftarrow$ product

SOP (sum of product) $\rightarrow A \oplus B = \bar{A}B + A\bar{B}$

Drawing $\bar{A}B + A\bar{B}$ which is XOR gate using basic gates.



$$A \oplus B = (A+B) \cdot (\bar{A} + \bar{B})$$

POS (product of sum)

0 → is barred

SOP

$$A \oplus B \oplus C = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + A\bar{B}C$$

POS → 1 → is barred.

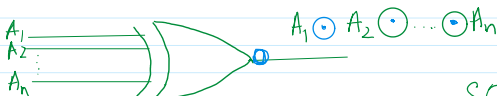
$$A \oplus B \oplus C = (A+B+C) \cdot (A+B+\bar{C}) \cdot (\bar{A}+B+\bar{C}) \cdot (\bar{A}+\bar{B}+C)$$

Home work, draw the logic ckt. by using basic gates.

XNOR:-

1. **XNOR Gate** : It takes 2 or more input signals and output one signal. The output signal is high or 1 or true, iff we had even number of high or 1 or true input signals, otherwise output is low or 0.

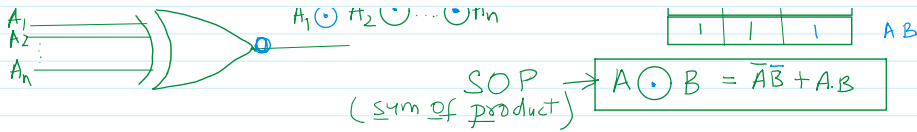
It is represented by '⊙' plus in a circle symbol.



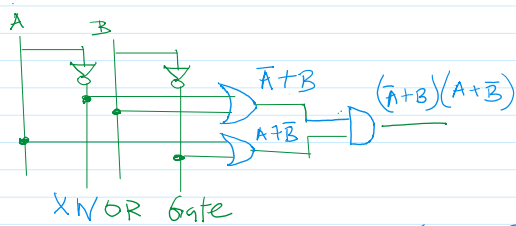
A	B	A ⊙ B
0	0	1
0	1	0
1	0	0
1	1	1

$\bar{A}\bar{B}$
 $(A+\bar{B})$
 $(\bar{A}+B)$
 AB } POS

SOP $\rightarrow A \odot B = \bar{A}\bar{B} + A.B$



POS $A \odot B = (A+B)(\bar{A}+\bar{B})$



H.W $\rightarrow$ Draw the ckt. using basic gates.

	A	B	C	A ⊙ B ⊙ C
0	0	0	0	1
1	0	0	1	0 ✓
2	0	1	0	0 ✓
3	0	1	1	1
4	1	0	0	0 ✓
5	1	0	1	1
6	1	1	0	1
7	1	1	1	0 ✓

$$G = A \odot B \odot C$$

SOP

$$G = \bar{A}\bar{B}\bar{C} + \bar{A}BC + A\bar{B}\bar{C} + ABC$$

POS

$$G = (A+B+C)(A+\bar{B}+\bar{C})(\bar{A}+B+\bar{C})(\bar{A}+\bar{B}+C)$$

H.W Draw the logic ckt using basic gates.

Next class:- Application

Boolean Algebraic Laws

1. Logical Addition (OR)

- $0 + 0 = 0$
- $0 + 1 = 1$
- $1 + 0 = 1$
- $1 + 1 = 1$

2. Logical Multiplication

- $0 \cdot 0 = 0$
- $0 \cdot 1 = 0$
- $1 \cdot 0 = 0$
- $1 \cdot 1 = 1$

3. Complement of bits

- $0' = 1$
- $1' = 0$

4. Laws of boolean algebra

Law	Properties
$0 + p = p$ $0 \cdot p = 0$	0
$1 + p = 1$ $1 \cdot p = p$	1
$p + p = p$ $p \cdot p = p$	Idempotence law
$p + p \cdot q = p$ $p + (p \cdot q) = p + q$	Absorption law
$(p')' = p$	Involution law
$p + p' = 1$ $p \cdot p' = 0$	Complementarity law
$(p + q) + r = p + (q + r)$ $(p \cdot q) \cdot r = p \cdot (q \cdot r)$	Associative law
$p(q+r) = pq + pr$ $p + q \cdot r = (p + q)(p + r)$ $p + p'q = p + q$	Distributive law
$(p + q)' = p' \cdot q'$ $(p \cdot q)' = p' + q'$	Demorgan's law
$(p \Rightarrow q) = p' + q$	Implication or

$$RHS \sim (p+q)(p+r)$$

$$= p \cdot p + p \cdot r + p \cdot q + q \cdot r$$

$$= \underline{p} + \underline{p \cdot r} + \underline{p \cdot q} + q \cdot r$$

$$= p(1+r+q) + q \cdot r$$

$$= p \cdot 1 + q \cdot r$$

$$= p + q \cdot r = LHS$$

$$LHS \sim p + p'q$$

$$= (p+p')(p+q) \text{ (using above)}$$

$$= \underline{1}(\underline{p+q})$$

	Conditional elimination or if...then
$(p \Rightarrow q) = (p \Rightarrow q) \cdot (q \Rightarrow p) = p' \cdot q' + p \cdot q$	Bi-conditional elimination or equivalence or if and only if (iff)

p	q	p'	q'	(p+q)'	p'·q'	(p·q)'	p' + q'
0	0	1	1	1	1	1	1
0	1	1	0	0	0	1	1
1	0	0	1	0	0	1	1
1	1	0	0	0	0	0	0

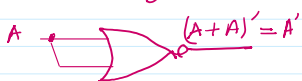
$$(p+q)' = p' \cdot q' \quad (p \cdot q)' = p' + q'$$

Demorgan's law is proved using truth table.

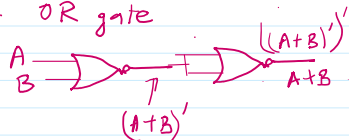
What is truth table?
A truth table is a systematic way to represent all possible combination of inputs and their corresponding outputs for a logical expression or circuit. It is widely used in Boolean Algebra, digital logic and design, and computer science to analyze and validate logical operations.

NOR to NOR logic: (Any types of basic gates using NOR gates only).

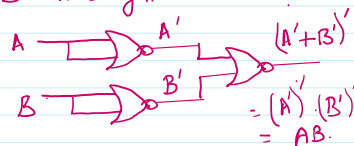
1. NOT gate



2. OR gate



3. AND gate



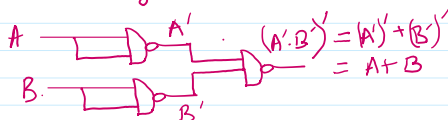
$$x^{1/10} + x^{1/20} + x^{1/30} = 56 \text{ find } x$$

NAND to NAND logic: (Any types of basic gates using NAND gates only).

1. NOT gate



2. OR gate



3. AND gate



Why NAND and NOR Gates are called Universal Gates?

Ans:- We can design ckts. using AND, OR and NOT gates but NAND and NOR gates are more popular as these are less expensive and easier to design. And, also other switching functions (AND, OR, NOT, XOR, XNOR) can be easily implemented using NAND/NOR gates.

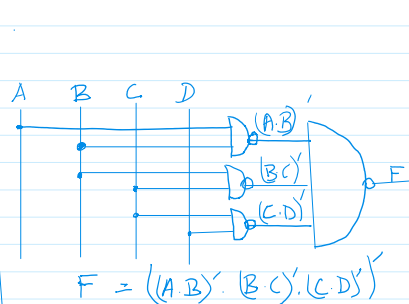
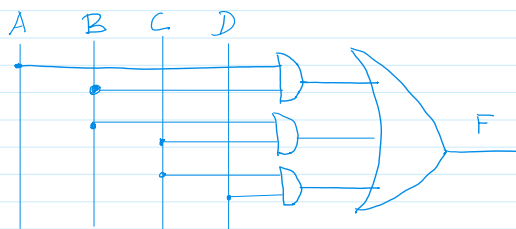
Thus NAND and NOR gates are also referred to as Universal gates.

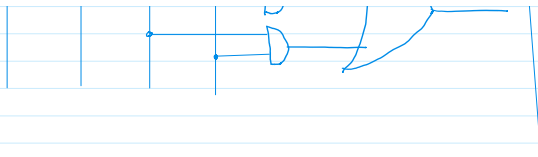
Skill:

Design rule for NAND to NAND logic network:

1. Draw simplified sum of product or SOP expression.
2. Draw the circuit diagram using AND OR NOT gates or basic gates.
3. Just replace AND OR NOT gates with NAND gates

$$F = \underline{A}B + B\underline{C} + C\underline{D}$$





$$F = (A \cdot B)' \cdot (B \cdot C)' \cdot (C \cdot D)'$$

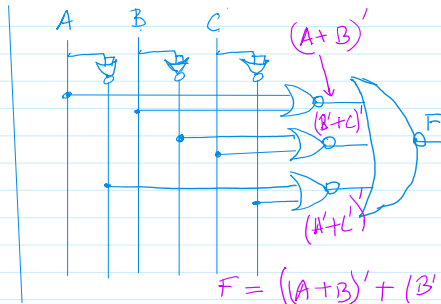
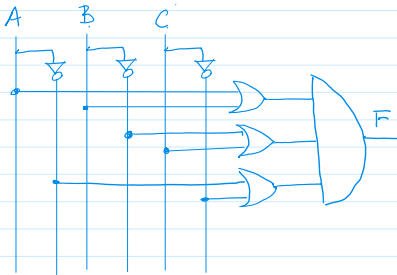
$$= ((A \cdot B)')' + ((B \cdot C)')' + ((C \cdot D)')'$$

$$= AB + BC + CD$$

Design rule for NOR to NOR logic network:

1. Draw simplified product of sum or POS expression.
2. Draw the circuit diagram using AND OR NOT gates or basic gates.
3. Just replace AND OR NOT gates with NOR gates

$$F = (A+B) \cdot (B'+C) \cdot (A'+C')$$



$$F = ((A+B)')' + ((B'+C)')' + ((A'+C')')'$$

$$= ((A+B)')' \cdot ((B'+C)')' \cdot ((A'+C')')'$$

$$= (A+B) \cdot (B'+C) \cdot (A'+C')$$

$$F = (A+B) \cdot (B'+C) \cdot (A'+C') \quad \text{Draw kkt. using NAND gates only.}$$

↓

$$\text{dual: } F = A \cdot B + B' \cdot C + A' \cdot C' \Rightarrow \text{then apply NAND to NAND logic.}$$

Home work- Draw XOR & XNOR circuits using.

1. NAND to NAND logic
2. NOR to NOR logic.